2025.11.26

Recall: $u \in L^1_{loc}(\Omega)$

$u_\varepsilon(x) = \int_\Omega j_\varepsilon(x-y)\, u(y)\, dy \quad \forall x \in \Omega_\varepsilon$

$\Omega_\varepsilon = \{x : dist(x, \partial\Omega) > \varepsilon\}$

Lemma1 $u \in L^1_{loc}(\Omega) \Rightarrow \forall \varepsilon > 0, u_\varepsilon \in C^\infty(\Omega)$

$(1 \le p \le \infty)\ u \in L^p(\Omega) \Rightarrow \forall \varepsilon > 0, u_\varepsilon \in C^\infty(\Omega)$

If Ω is bdd, $u_\varepsilon \in C_c^\infty(\Omega)$

Lemma2 $u \in C(\Omega)$ $\Omega' \subset\subset \Omega$

$\Rightarrow u_\varepsilon \to u$ in $C(\overline{\Omega'})$

Lemma3 $u \in L^1_{loc}(\Omega) \Rightarrow u_\varepsilon \to u$ a.e.

Lemma4 $u \in L^p_{loc}(\Omega)$ $(L^p(\Omega))$

$(1 \le p < \infty) \Rightarrow u_\varepsilon \to u$ (L^p_{loc} / L^p)

Application: $1 \le p < \infty$

(1) $L^p(\Omega) \cap C^\infty(\Omega)$ is dense in L^p

(Lemma1 + Lemma4)

(2) Ω is bdd, Then

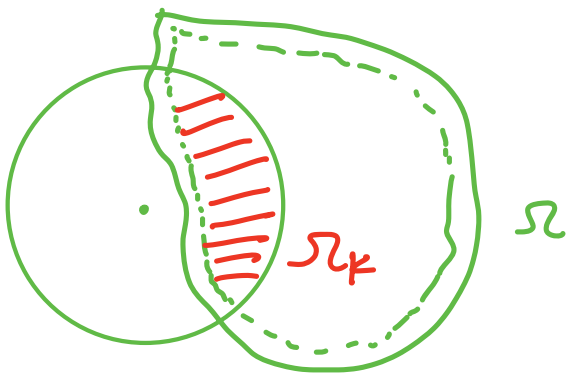
(Lemma1 + Lemma4) $\Rightarrow$

C^∞ is dense in $L^p(\Omega)$

(3) $C_c^\infty(\Omega)$ is dense in $L^p(\Omega)$

Pf: $\forall u \in L^p(\Omega)$. Define

$\Omega_k = \{x \in \Omega : |x| < k, dist(x, \partial\Omega) > \frac{1}{k}\}$



$\Omega_k \subset\subset \Omega_{k+1} \subset\subset \Omega \Rightarrow \bigcup_{k=1}^\infty \Omega_k = \Omega$.

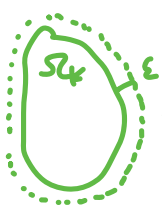
Let $u_k(x) = u(x) \mathbb{1}_{\Omega_k}(x) \in L^p(\Omega)$

$\Rightarrow u_k(x) \to u(x)$ in $L^p(\Omega)$

For fixed $k \ge 1$: $(u_k)_\varepsilon \xrightarrow{\varepsilon \to 0} u_k$ in $L^p(\Omega)$

$supp(u_k)_\varepsilon$ & $(u_k)_\varepsilon \in C^\infty(\mathbb{R}^n)$

$\Rightarrow (u_k)_\varepsilon \subset C_c^\infty(\Omega)$ ε small



$\forall \delta > 0, \exists k$ s.t. $\|u_k - u\|_{L^p(\Omega)} < \frac{\delta}{2}$

Take $\varepsilon>0$ small s.t. $\|(u_k)_\varepsilon - u_k\|_{L^p} < \frac{\delta}{2}$. Thus

$$\|(u_k)_\varepsilon - u\|_{L^p} \le \|(u_k)_\varepsilon - u_k\|_{L^p} + \|u_k - u\|_{L^p} < \delta$$

(4) $u \in L^1_{loc}(\Omega)$, $v_1, v_2 \in L^1_{loc}(\Omega)$ are α-th weak derivatives of u then $v_1 = v_2$ a.e. (Uniqueness of weak derivative)

pf: $\forall \varphi \in C_c^\infty(\Omega) \Rightarrow$

$$\int_\Omega v_1 \varphi\, dx = \int_\Omega v_2 \varphi\, dx = (-1)^{|\alpha|}\int_\Omega u \partial^\alpha \varphi\, dx$$

$$\Rightarrow \int_\Omega (v_1 - v_2)\varphi\, dx = 0$$

$\forall$ fixed $y \in \Omega$, take $\varphi(x) = j_\varepsilon(y-x)$

$\operatorname{supp}\varphi = B_\varepsilon(y) \subset\subset \Omega$ if $\varepsilon>0$ small.

$$\Rightarrow \int_\Omega (v_1 - v_2) j_\varepsilon(y-x)\, dx$$

By Lemma 4, $v_\varepsilon \to v_1 - v_2$ in $L^1_{loc}(\Omega)$, $\|\cdot\| \to 0$

$\Rightarrow v_1 = v_2$ a.e. $x \in \Omega$.

By product, $\forall\, v \in L^1_{loc}(\Omega)$ if $\int_\Omega v\cdot\varphi\, dx = 0$ $\forall \varphi \in C_c^\infty(\Omega)$

$\Rightarrow v = 0$ a.e. $x \in \Omega$.

Lemma 5 If $u \in L^1_{loc}(\Omega)$, weak $\partial^\alpha u$ exist $[\exists\, v \in L^1_{loc}(\Omega)$ $\partial^\alpha u = v$ in sense of distribution]

then $\forall \varepsilon>0$, $\forall x \in \Omega_\varepsilon$, classical $\partial^\alpha u_\varepsilon(x) = (\partial^\alpha u)_\varepsilon(x)$

pf: $\forall \varepsilon>0$, $\forall x \in \Omega_\varepsilon$, $u_\varepsilon(x) = \int_\Omega j_\varepsilon(x-y)u(y)\, dy$

$$\partial^\alpha u_\varepsilon(x) = \int_\Omega \partial_x^\alpha j_\varepsilon(x-y)\, u(y)\, dy$$

$$= (-1)^{|\alpha|}\int_\Omega (\partial_y^\alpha j_\varepsilon(x-y))\, u(y)\, dy$$

(weak derivative)

$j_\varepsilon(x-\cdot) \in C_c^\infty(\Omega)$ $= \int_\Omega j_\varepsilon(x-y)\, \partial_y^\alpha u(y)\, dy$

$= (\partial^\alpha u)_\varepsilon$ not hold

Q: If u, weak $\partial^\alpha u \in L^1(\Omega)$, does $(\partial^\alpha u)_\varepsilon = \partial^\alpha u_\varepsilon$ for all $x \in \Omega$.

In general it may not for $x \in \Omega \setminus \Omega_\varepsilon$.

Since $j_\varepsilon(x-\cdot) \notin C_c^\infty(\Omega)$

However, if $\operatorname{supp} u \subset\subset \Omega$, then $\partial^\alpha u_\varepsilon = (\partial^\alpha u)_\varepsilon$ on Ω provided $\varepsilon>0$ small

(5) Suppose $u \in L^1_{loc}(\Omega)$, weak $\frac{\partial u}{\partial x_i}$ $i=1\cdots n$ exist $\in L^1_{loc}(\Omega)$, weak $\nabla u = 0$ a.e. Ω

Then $u \equiv$ constant C on Ω.

pf: By Lemma 5, $\nabla u_\varepsilon = (\nabla u)_\varepsilon$ in Ω_ε

$\Rightarrow \nabla u_\varepsilon = 0$, $u_\varepsilon \equiv C_\varepsilon$ in Ω_ε

By Lemma 3, $u_\varepsilon \to u$ a.e.

$\Rightarrow u \equiv C$ a.e. in Ω.

Thm 1 Let $u, v \in L^1_{loc}(\Omega)$, then

$v =$ weak $\partial^\alpha u \Leftrightarrow \exists \{u_k\}_{k=1}^\infty \subset C^\infty(\Omega)$ s.t.

$u_k \to u$ in $L^1_{loc}(\Omega)$ (*1) classical:

$\partial^\alpha u_k \to v$ in $L^1_{loc}(\Omega)$ (*2)

pf: ($\Leftarrow$): $\forall \varphi \in C_0^\infty(\Omega)$

$$\int_\Omega (\partial^\alpha u_k)\varphi\, dx = (-1)^{|\alpha|}\int_\Omega u_k \partial^\alpha \varphi\, dx$$

$\downarrow$(*2) $\qquad\qquad$ $\downarrow$(*1)

$$\int_\Omega v\varphi\, dx = (-1)^{|\alpha|}\int_\Omega u\partial^\alpha\varphi\, dx = \int_\Omega \partial^\alpha u\varphi\, dx$$

$\Rightarrow v = \text{weak } \partial^\alpha u$.

($\Rightarrow$): $\forall k \geq 1$, Let

$\Omega_k = \{x \in \Omega : dist(x, \partial\Omega) > \frac{1}{k}\}$

$\Rightarrow \Omega_k \subset\subset \Omega$ & $\Omega_k \uparrow \Omega$

Let $u_k(x) = \int_{\Omega_k} j_{\frac{1}{k}}(x-y)u(y)dy$

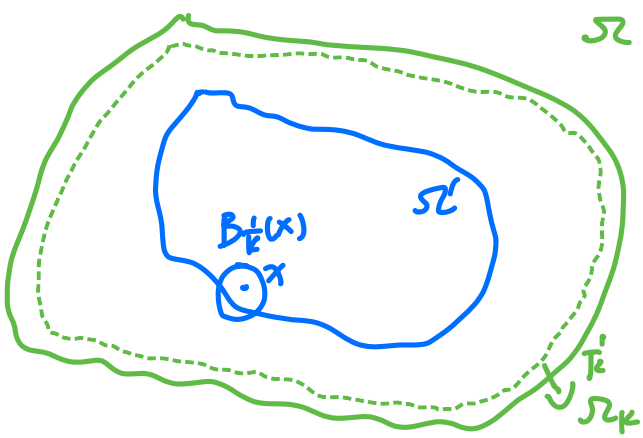
$= (u|_{\Omega_k})_{\varepsilon=\frac{1}{k}}(x)$

$u|_{\Omega_k} \in L^1(\Omega_k)$. By Lemma1, $\Rightarrow$

$u_k(x) \in C_c^\infty(\Omega) \subset C^\infty(\Omega)$

For fixed $\Omega' \subset\subset \Omega$, W.T.S.:

$$\begin{cases} u_k \to u \text{ in } L^1(\Omega') \\ \partial^\alpha u_k \to \partial^\alpha u \text{ in } L^1(\Omega') \end{cases} \quad k \to \infty$$



$\exists k_0 \geq 1$, s.t. when $k \geq k_0$, $\Omega' \subset\subset \Omega_k$.

$\forall x \in \Omega'$, $u_k(x) = \int_{\Omega_k} j_{\frac{1}{k}}(x-y)u(y)dy$

take k_0 large s.t. $B_{\frac{1}{k}}(x) \subset\subset \Omega_{k_0}$

$\Omega_k \leftrightarrow \Omega_k \cap B_{\frac{1}{k}}(x)$. Hence,

$u_k(x) = \int_{\Omega_{k_0}} j_{\frac{1}{k}}(x-y)u(y)dy$

$= (u|_{\Omega_{k_0}})_{\varepsilon=\frac{1}{k}}(x) \in L^1(\Omega_{k_0})$

By Lemma4, $u_k(x) \to u|_{\Omega_{k_0}}$ in $L^1(\Omega_{k_0})$

$\Leftrightarrow u_k(x) \to u(x)$ in $L^1(\Omega')$

By Lemma5.

$\partial^\alpha u_k(x) = \partial^\alpha (u|_{\Omega_k})_{\varepsilon=\frac{1}{k}}(x) = (\partial^\alpha u|_{\Omega_k})_{\frac{1}{k}}$

$= (v|_{\Omega_k})_{\frac{1}{k}}$ By Lemma4, Let $k \to \infty$,

$\partial^\alpha u_k \to v|_{\Omega_k}$ in $L^1(\Omega')$

$\forall x \in (\Omega_{k_0})_{\frac{1}{k}} \supset \Omega'$